Cyberbullying Detection using Machine Learning

MCA Final Year Project

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Department: Master of Computer Applications

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Import Required Libraries

In [77]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import nltk
import re
import joblib

from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.naive_bayes import MultinomialNB
from sklearn.svm import LinearSVC
from sklearn.multiclass import OneVsRestClassifier
from sklearn.metrics import classification_report, f1_score
```

Download NLTK Resources

In [78]:
```
# %%
nltk.download('stopwords')
nltk.download('wordnet')
```

```
[nltk_data] Downloading package stopwords to
[nltk_data]     C:\Users\anshu\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data]     C:\Users\anshu\AppData\Roaming\nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
```

Out[78]: True

Load Dataset

In [79]:
```
df = pd.read_csv("data/train.csv")

label_columns = ['toxic','severe_toxic','obscene','threat','insult','identity_ha
```

```
print("Dataset Shape:", df.shape)
print("\nMissing Values:\n", df.isnull().sum())
df.head()
```

```
Dataset Shape: (159571, 8)

Missing Values:
 id               0
comment_text     0
toxic            0
severe_toxic     0
obscene          0
threat           0
insult           0
identity_hate    0
dtype: int64
```

Out[79]:

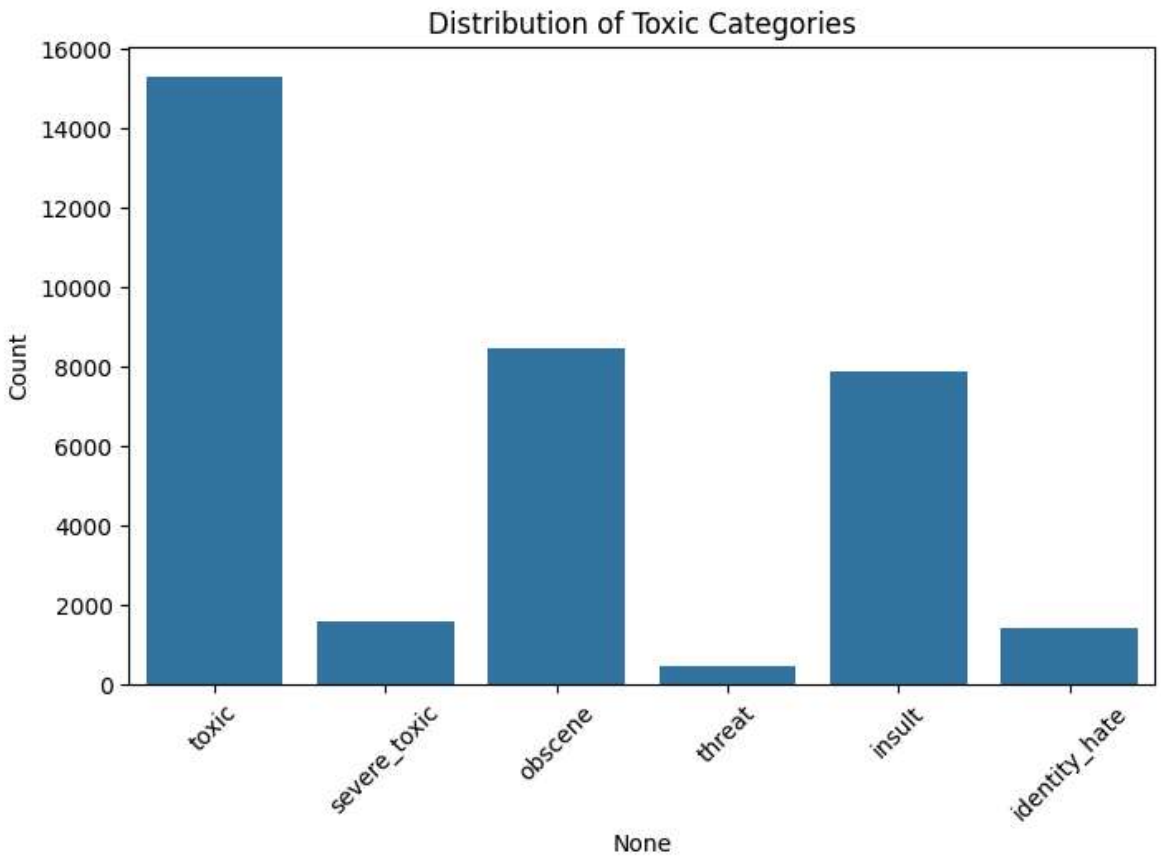
	id	comment_text	toxic	severe_toxic	obscene	threat	insult	ide
0	0000997932d777bf	Explanation\nWhy the edits made under my usern...	0	0	0	0	0	
1	000103f0d9cfb60f	D'aww! He matches this background colour I'm s...	0	0	0	0	0	
2	000113f07ec002fd	Hey man, I'm really not trying to edit war. It...	0	0	0	0	0	
3	0001b41b1c6bb37e	"\nMore\nI can't make any real suggestions on ...	0	0	0	0	0	
4	0001d958c54c6e35	You, sir, are my hero. Any chance you remember...	0	0	0	0	0	

Exploratory Data Analysis

In [80]:

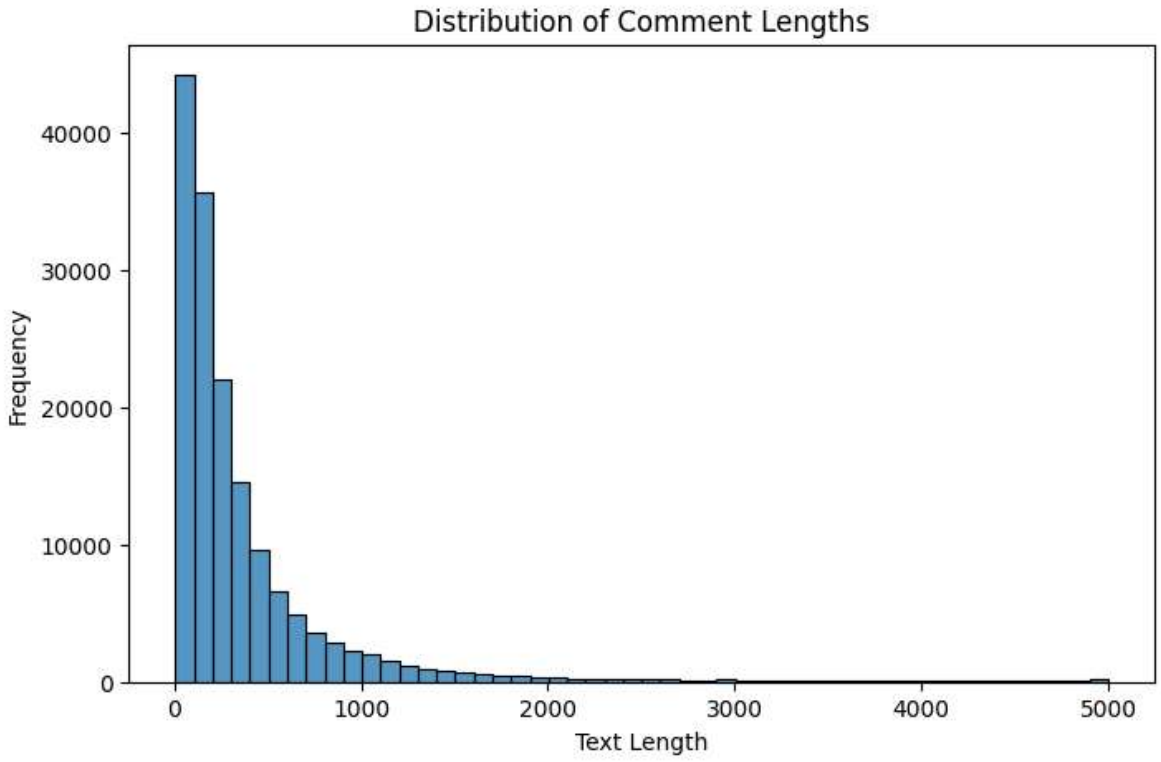
```
label_counts = df[label_columns].sum()

plt.figure(figsize=(8,5))
sns.barplot(x=label_counts.index, y=label_counts.values)
plt.title("Distribution of Toxic Categories")
plt.xticks(rotation=45)
plt.ylabel("Count")
plt.show()
```



In [81]:
```
df['text_length'] = df['comment_text'].apply(len)

plt.figure(figsize=(8,5))
sns.histplot(df['text_length'], bins=50)
plt.title("Distribution of Comment Lengths")
plt.xlabel("Text Length")
plt.ylabel("Frequency")
plt.show()
```



Text Preprocessing

In [82]:
```
lemmatizer = WordNetLemmatizer()
stop_words = set(stopwords.words('english'))

def clean_text(text):
    text = text.lower()
    text = re.sub(r'http\S+', '', text)
    text = re.sub(r'[^a-zA-Z]', ' ', text)
    words = text.split()
    words = [lemmatizer.lemmatize(word) for word in words if word not in stop_wo
    return " ".join(words)

df['clean_text'] = df['comment_text'].apply(clean_text)
df[['comment_text', 'clean_text']].head()
```

Out[82]:

	comment_text	clean_text
0	Explanation\nWhy the edits made under my usern...	explanation edits made username hardcore metal...
1	D'aww! He matches this background colour I'm s...	aww match background colour seemingly stuck th...
2	Hey man, I'm really not trying to edit war. It...	hey man really trying edit war guy constantly ...
3	"\nMore\nI can't make any real suggestions on ...	make real suggestion improvement wondered sect...
4	You, sir, are my hero. Any chance you remember...	sir hero chance remember page

TF-IDF Feature Engineering

In [83]:
```
tfidf = TfidfVectorizer(max_features=5000)
X = tfidf.fit_transform(df['clean_text'])
y = df[label_columns]

print("Feature Matrix Shape:", X.shape)
print("Target Shape:", y.shape)
```

```
Feature Matrix Shape: (159571, 5000)
Target Shape: (159571, 6)
```

Train-Test Split

In [84]:
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("Training Shape:", X_train.shape)
print("Testing Shape:", X_test.shape)
```

```
Training Shape: (127656, 5000)
Testing Shape: (31915, 5000)
```

Multi-Label Model Training & Comparison

In [85]:
```
# Logistic Regression
log_model = OneVsRestClassifier(LogisticRegression(max_iter=1000))
log_model.fit(X_train, y_train)
y_pred_log = log_model.predict(X_test)

print("\n===== Logistic Regression =====")
print(classification_report(y_test, y_pred_log, target_names=label_columns))
log_f1 = f1_score(y_test, y_pred_log, average='macro')
```

```
===== Logistic Regression =====
               precision    recall  f1-score   support

        toxic       0.91      0.63      0.74      3056
 severe_toxic       0.60      0.26      0.37       321
      obscene       0.91      0.65      0.76      1715
       threat       0.67      0.14      0.22        74
       insult       0.82      0.53      0.65      1614
identity_hate       0.74      0.18      0.29       294

    micro avg       0.88      0.57      0.69      7074
    macro avg       0.77      0.40      0.51      7074
 weighted avg       0.87      0.57      0.68      7074
  samples avg       0.06      0.05      0.05      7074
```

```
d:\Code X\Projects\iter final year\Cyber Bullying\.zenv\Lib\site-packages\sklearn
\metrics\_classification.py:1833: UndefinedMetricWarning: Precision is ill-define
d and being set to 0.0 in samples with no predicted labels. Use `zero_division` p
arameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
d:\Code X\Projects\iter final year\Cyber Bullying\.zenv\Lib\site-packages\sklearn
\metrics\_classification.py:1833: UndefinedMetricWarning: Recall is ill-defined a
nd being set to 0.0 in samples with no true labels. Use `zero_division` parameter
to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
d:\Code X\Projects\iter final year\Cyber Bullying\.zenv\Lib\site-packages\sklearn
\metrics\_classification.py:1833: UndefinedMetricWarning: F-score is ill-defined
and being set to 0.0 in samples with no true nor predicted labels. Use `zero_divi
sion` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
```

In [86]:
```
# Naive Bayes
nb_model = OneVsRestClassifier(MultinomialNB())
nb_model.fit(X_train, y_train)
y_pred_nb = nb_model.predict(X_test)

print("\n===== Naive Bayes =====")
print(classification_report(y_test, y_pred_nb, target_names=label_columns))
nb_f1 = f1_score(y_test, y_pred_nb, average='macro')
```

```
===== Naive Bayes =====
               precision    recall  f1-score   support

        toxic       0.92      0.53      0.67      3056
 severe_toxic       0.60      0.18      0.28       321
      obscene       0.90      0.55      0.68      1715
       threat       0.00      0.00      0.00        74
       insult       0.82      0.46      0.59      1614
identity_hate       0.63      0.07      0.13       294

    micro avg       0.88      0.48      0.62      7074
    macro avg       0.65      0.30      0.39      7074
 weighted avg       0.86      0.48      0.61      7074
  samples avg       0.05      0.04      0.04      7074
```

```
d:\Code X\Projects\iter final year\Cyber Bullying\.zenv\Lib\site-packages\sklearn
\metrics\_classification.py:1833: UndefinedMetricWarning: Precision is ill-define
d and being set to 0.0 in labels with no predicted samples. Use `zero_division` p
arameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
d:\Code X\Projects\iter final year\Cyber Bullying\.zenv\Lib\site-packages\sklearn
\metrics\_classification.py:1833: UndefinedMetricWarning: Precision is ill-define
d and being set to 0.0 in samples with no predicted labels. Use `zero_division` p
arameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
d:\Code X\Projects\iter final year\Cyber Bullying\.zenv\Lib\site-packages\sklearn
\metrics\_classification.py:1833: UndefinedMetricWarning: Recall is ill-defined a
nd being set to 0.0 in samples with no true labels. Use `zero_division` parameter
to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
d:\Code X\Projects\iter final year\Cyber Bullying\.zenv\Lib\site-packages\sklearn
\metrics\_classification.py:1833: UndefinedMetricWarning: F-score is ill-defined
and being set to 0.0 in samples with no true nor predicted labels. Use `zero_divi
sion` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
```

In [87]:

```python
# Linear SVM
svm_model = OneVsRestClassifier(LinearSVC())
svm_model.fit(X_train, y_train)
y_pred_svm = svm_model.predict(X_test)

print("\n===== Linear SVM =====")
print(classification_report(y_test, y_pred_svm, target_names=label_columns))
svm_f1 = f1_score(y_test, y_pred_svm, average='macro')
```

```
===== Linear SVM =====
               precision    recall  f1-score   support

        toxic       0.88      0.68      0.77      3056
 severe_toxic       0.56      0.26      0.35       321
      obscene       0.89      0.69      0.78      1715
       threat       0.62      0.20      0.31        74
       insult       0.80      0.58      0.67      1614
identity_hate       0.71      0.26      0.38       294

    micro avg       0.85      0.62      0.72      7074
    macro avg       0.74      0.45      0.54      7074
 weighted avg       0.84      0.62      0.71      7074
  samples avg       0.06      0.06      0.06      7074
```

```
d:\Code X\Projects\iter final year\Cyber Bullying\.zenv\Lib\site-packages\sklearn
\metrics\_classification.py:1833: UndefinedMetricWarning: Precision is ill-define
d and being set to 0.0 in samples with no predicted labels. Use `zero_division` p
arameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
d:\Code X\Projects\iter final year\Cyber Bullying\.zenv\Lib\site-packages\sklearn
\metrics\_classification.py:1833: UndefinedMetricWarning: Recall is ill-defined a
nd being set to 0.0 in samples with no true labels. Use `zero_division` parameter
to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
d:\Code X\Projects\iter final year\Cyber Bullying\.zenv\Lib\site-packages\sklearn
\metrics\_classification.py:1833: UndefinedMetricWarning: F-score is ill-defined
and being set to 0.0 in samples with no true nor predicted labels. Use `zero_divi
sion` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
```

In [88]:

```python
print("\n===== Model Comparison (Macro F1 Score) =====")
print("Logistic Regression F1:", round(log_f1,4))
print("Naive Bayes F1:", round(nb_f1,4))
print("Linear SVM F1:", round(svm_f1,4))
```

```
===== Model Comparison (Macro F1 Score) =====
Logistic Regression F1: 0.5058
Naive Bayes F1: 0.3924
Linear SVM F1: 0.5427
```

Select and Save Best Model

In [89]:

```python
import os
os.makedirs("models", exist_ok=True)

models = {
    "Logistic Regression": (log_f1, log_model),
    "Naive Bayes": (nb_f1, nb_model),
    "Linear SVM": (svm_f1, svm_model)
}

best_model_name, (best_f1, best_model) = max(
    models.items(),
    key=lambda x: x[1][0]
)

joblib.dump(best_model, "models/best_multilabel_model.pkl")
joblib.dump(tfidf, "models/tfidf_vectorizer.pkl")

print(f"Best model selected: {best_model_name}")
print(f"Macro F1 Score: {round(best_f1,4)}")
print("Model and vectorizer saved successfully.")
```

```
Best model selected: Linear SVM
Macro F1 Score: 0.5427
Model and vectorizer saved successfully.
```

Prediction Function

Test

In [90]:
```
def predict_multilabel(text):
    cleaned = clean_text(text)
    vectorized = tfidf.transform([cleaned])
    prediction = best_model.predict(vectorized)[0]

    detected_labels = []

    for label, value in zip(label_columns, prediction):
        if value == 1:
            detected_labels.append(label.replace("_", " ").title())

    print("\n========================================")
    print("Input Text:")
    print(text)
    print("----------------------------------------")

    if detected_labels:
        print("⚠ Toxic Categories Detected:")
        for label in detected_labels:
            print(f"  • {label}")
        print(f"\nTotal Categories Detected: {len(detected_labels)}")
    else:
        print("✅ No Toxic Content Detected")

    print("========================================\n")

    return detected_labels  # <- Now it returns something
```

In [92]:
```
predict_multilabel("You are such an idiot.")
predict_multilabel("Thank you for your help.")
predict_multilabel("I will kill you .")
```

```
========================================
Input Text:
You are such an idiot.
----------------------------------------
⚠️ Toxic Categories Detected:
  • Toxic
  • Obscene
  • Insult

Total Categories Detected: 3
========================================


========================================
Input Text:
Thank you for your help.
----------------------------------------
✅ No Toxic Content Detected
========================================


========================================
Input Text:
I will kill you .
----------------------------------------
⚠️ Toxic Categories Detected:
  • Toxic
  • Threat

Total Categories Detected: 2
========================================
```

Out[92]: ['Toxic', 'Threat']